

POAMS: LECTURE 10
THE ULTIMATELY DISCRETE NATURE OF
PHYSICAL PHENOMENA

Having presented our Normal Realist approach to Special Relativity, I now turn to questions regarding physics at the micro-physical level. In particular, I will be concerned with the ultimately discrete nature of physical phenomena and the origins of Quantum Mechanics. This lecture is based on some of 'Light Speed...' Chapter 5.

1. The quantisation of physical measures

Relativity, as is science in general, is primarily concerned with *observation*, which is essentially *information*. But such communication of information is impossible where its elements are all determinately connected-up in some geometrical four-dimensional continuum, as Einstein envisaged. For observation to be real, communication of information has to be real; and information, as Claude Shannon (1916-2001) and Warren Weaver (1894-1978) demonstrated in their pioneering work on *Communication Theory* in 1948, is a measure of *improbability*. But there can be no improbability in sequences of events that are either mechanically or geometrically determined. Such sequences are therefore devoid of informational content.

A 'relativism' like Einstein's then, which assumes that all observational event sequences are geometrically connected-up in some continuous 'substratum' is philosophically incoherent, which goes a long way towards Mach's rejection of Einstein's Relativity. In true relativism all observational events have to be *discrete* and indeterminate. Like characters in a printer's font, they cannot convey information without the efforts of a compositor (*i.e.*, an observer). If those characters were all linked together, the compositor's work would be impossible. [*Continuous information cannot be recorded in any meaningful way.*]

In contrast, to Einstein's Relativity, from the Normal Realist standpoint, since all fundamental physical quantities are based on observations, it follows that on a small enough scale, they must be ultimately *discrete*, rather than continuous as in classical physics.

The discrete nature of physical measures is also manifest on the macro-scale, albeit in a different way. For example, although, mathematically speaking, the *irrational* numbers provide the completion of the set of *rational* numbers (quotients of integers) to the set of *real* numbers, thus providing a *continuum* on which to build calculus, such irrational numbers are not recordable in any sense of practical measurement. Any measuring device can only record rational approximations of irrational numbers. All recordable (empirical) physical measures are therefore, de facto, ultimately *discrete*.

This fundamental *discretisation i.e. quantisation* of physical phenomena was first suggested by the studies of Max Planck (1858-1947). This led to the beginnings of *Quantum Mechanics*, to be described shortly. Although all physical measures are ultimately discrete on a small enough scale, the fact that these are so small makes them appear, for all practical purposes, to be continuous on the macro-scale. This is known as the *correspondence principle* in Quantum Mechanics. Indeed, in order to construct a *mathematical model* of some physical phenomenon on the macro-scale, as opposed to the phenomenon itself, we must necessarily use continuous variables.

The fact that physical measures are ultimately *quantised, i.e.* occur in discrete ‘packets’, leads us to consider the fundamental difference, if any, between physical phenomena on the macro- and micro-levels. This consideration leads to two fundamentally different approaches to physics. Classical physics is what might be described as an *analog* approach, in which the explanations of physical phenomena are couched in terms of conceptions of mechanical processes of essentially the same kind as those we observe on levels of ordinary sensation and perception. In that way of thinking, the ultimate elements of things are microphysical bits of matter moving about in essentially the same space as that in which things move about on ordinary macro-physical levels. By contrast, what we may call a *digital* approach to physics is one which seeks to explain all physical phenomena in terms of mathematical systems of discrete elements, which, in themselves, have none of the usual physical features of motion, duration, *etc.*, being the ultimately informational *bits* into which all those phenomena are analysed.

However, there are two kinds of these digital approaches which need logically to be distinguished. The aim of the first is to produce explanations in terms of elements such as pure numbers (for example, digital matrices of abstract and intrinsically featureless ones and noughts), whereas the aim of the second is to explain things in terms of systems of actual sensory/instrumental information (what Mach termed ‘sense-data’ [see *Lecture 3*]). In the first of these approaches, which we may call the *numerical approach*, the ‘geometry’ of the things we see in the world is an automatic construct out of relations between pure numbers, in the manner conceived originally by the Pythagoreans. Our Normal Realist philosophy, however, dictates that we follow the second digital approach, in which geometry is essentially a natural *a priori* feature of *phenomena*, the ultimate elements of which are the end-result of a reductive process of analysis of such phenomena, terminating, empirically, in further-irreducible quantum measures of physical *information*. We call this approach the *sense-datum approach*.

Now a flow of real information that does not involve a transfer of energy is inconceivable. Indeed, since the pioneering work of Shannon and Weaver, it is known that both information and energy are governed by virtually the same law, namely the statistical *Second Law of Thermodynamics*. A quantum of observational information is therefore also a quantum of energy, and a change in energy, in the form of gain or loss, is the product of energy and time, known as **action**. So any natural limit to the reduction of observational information, *i.e.*, ultimately light, implies a corresponding limit to the reduction of action. If there is an empirical limit of reduction of action/information flow, then any increase in that that quantity has to be in integer multiplies of that fundamental unit. In that absolute limit, any further reduction of the components of action, *i.e.*, energy and time, can take place only reciprocally. That is to say, any reduction in the energy can be achieved only at the cost of increasing the time, and vice-versa, so that the fundamental unit of action remains the same. I will now show that there is, in fact, such a natural limit of reduction of action in Quantum Mechanics, of just the sort to be expected on Normal Realist grounds.

2. The origins of Quantum Mechanics

The beginning of Quantum Mechanics can be traced back to Max Planck's proposal that energy is *quantised*. This was to explain the apparently paradoxical results of the Rayleigh-Jeans study of *black body* radiation carried out by Lord Rayleigh (1842-1919) and James Jeans (1877-1946). In this context, a *black body* is an object which absorbs and then re-emits all electromagnetic radiation that falls onto it. In 1900, Rayleigh and Jeans studied the problem of the radiant energy confined to a container whose walls are *black* in this sense. Using classical physics, they derived a formula for the energy per unit volume of radiated electromagnetic waves of a certain frequency. This depended only on the frequency and the temperature of the enclosure. Their formula not only firmly contradicted results of experiments for high frequencies but also predicted that the total energy of radiation per unit volume of the container would be infinite! This result became known as the *ultraviolet catastrophe*.

Planck provided an answer to this dilemma about a year later. Rayleigh and Jeans had supposed that the radiant energy was absorbed and re-emitted by the black body in a classical *continuous* way. Planck, on the other hand, proposed that this radiant energy could be emitted and absorbed only in the form of discrete 'packets', which he called **quanta**. According to Planck, electromagnetic radiation with *frequency* ν (*i.e.* oscillating at ν times per second) is made up of a *whole number* of quanta of energy, E , proportional to ν , so that

$$E = h\nu \quad (1)$$

where h is a constant known as *Planck's constant*. In other words, electromagnetic radiation is *ultimately discrete*, in the same way that a bowl of sugar consists of individual grains. Any electromagnetic radiation with frequency ν is then an integer multiple of E given by equation (1). Notice that, here, Planck's constant is the irreducible amount of *action*, talked of earlier.

Using this postulate, Planck replaced the Rayleigh-Jeans formula with a new one known as *Planck's Law*. Not only did this new formula agree entirely with the results of experiments and with the Rayleigh-Jeans formula for low frequencies, it also avoided the ultraviolet catastrophe. He

was able to deduce the value of h using the known values of observed frequencies, temperature and total energies and obtained

$$h \cong 6.626 \times 10^{-34} \text{ Joule seconds} \quad (2)$$

Of course, this value of h is tiny, so that on the macro-scale, the discrete energy levels become so close that the radiant energy appears to be continuous. This is an example of the correspondence principle in quantum physics mentioned earlier.

In 1905, Einstein called upon Planck's ideas to explain the *photoelectric effect*, which concerned the way in which 'electrons' were ejected from metals by incident light. According to the firmly accepted *wave theory* of light, discussed in Lecture 4, the number of electrons ejected could not depend on the frequency of the light. However, experiments showed that they did. Such experiments demonstrated that below a certain critical frequency, no electrons were ejected, regardless of the intensity of the radiation. As a solution to this dilemma, Einstein suggested that not only is energy exchanged between matter and radiation in discrete packets, but also that electromagnetic radiation itself actually consists of *action quanta*, h , which he called *photons*, each of energy given by equation (1). Below the critical frequency, he argued, the energy of the photon would be too weak to displace an electron. This appeared to be confirmation, by modern physics, of Newton's once prestigious 'corpuscular' or *particle theory* of light.

3. The Balmer-Rydberg formula and Bohr's model for the atom.

Another major contribution to the development of Quantum Mechanics involved experimental spectroscopy. In 1885, a Swiss Secondary School teacher, Johann Balmer (1825-1898), observed that the spectrum of light emitted by hydrogen gas forms an informational structure very similar to what we would now call a 'bar-code'. He was the first person to produce a successful mathematical formula for the distribution of these spectral lines. Then, in 1890, a Swedish physicist, Johannes Rydberg (1854-1919), extended Balmer's work and discovered a general rule applicable to the spectral lines of many other elements. Stated in terms of the frequency of the light, the *generalised Balmer-Rydberg formula* may be expressed as

$$\nu(n_1, n_2) = c\alpha_N R N^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (3)$$

In this formula, ν is the frequency of the light in cycles per second, N is the atomic number of the element (*e.g.*, $N = 1$ for hydrogen), R is a constant, known as the *Rydberg constant*, n_1 and n_2 are natural (whole) numbers representing the fixed and running terms respectively, and α_N is a constant depending on the element, with $\alpha_N \cong 1$. In Balmer's original formula, $N = 1$, $\alpha_N = 1$ and $n_1 = 2$. For $n_1 = 1$, the formula gives the *Lyman series* and $n_1 = 3$ gives the *Paschen series*.

The formula (3) was obtained *empirically* by Balmer and Rydberg, so as to fit the observed spectral lines. They offered no underlying *theory* as to *why* this formula might be true. It was not until 1913 that the first theoretical explanation emerged. This was proposed by Niels Bohr, who based his derivation of the formula on a mixture of classical mechanics and electrodynamical theory, plus the inclusion of the *quantisation of angular momentum*.

The optical spectral lines, Bohr supposed, were energies released by the elementary 'charged' particles known as *electrons*, orbiting around the *nucleus* of the atom, as in the Rutherford model of the atom developed in 1911 by Ernest Rutherford (1871-1937). But rather than having a continuous motion, as in Rutherford's model, Bohr supposed that these electrons changed orbits in discrete quantum jumps of energy, given by equation (1). In Bohr's model for the atom, he supposed that the electrons were confined to certain discrete orbits only. More precisely, he supposed that there exists a discrete set of stable states of an atom and transitions occur from one of these states to another with the emission of radiation. If the initial and final states have energies E_n and E_m , then the frequency of the emitted radiation is, by equation (1), $(E_n - E_m)/h$.

For simplicity, Bohr considered the electron orbits to be *circular*. He postulated that the magnitude of the angular momentum of the electrons in these orbits was quantised, in discrete units of $\hbar = h/2\pi$ since the circumference of a circle of unit radius is 2π . Hence, whereas h is the quantum of radiant energy as Einstein saw it, $\hbar$ is the quantum of *angular momentum* in Bohr's model. Angular momentum will be discussed at

length in later lectures, but for now it is sufficient to note that the **magnitude** of the **angular momentum** of a particle of mass m , following a circular orbit of radius r about the centre at speed v is mvr . Bohr therefore postulated that the electron of mass m and speed v could occupy only a circular orbit of radius r which satisfied

$$mvr = \frac{nh}{2\pi} \quad (4)$$

where n is a natural (whole) number, i.e., $n = 1, 2, 3, \dots$

We now specifically consider Bohr's model of the *hydrogen atom*. He supposed that the single electron to be moving in a circular orbit of radius r and with velocity v . He supposed that there was a 'charge', e , on the single *proton* in the nucleus and that the electron had a corresponding 'charge' $-e$. Classically speaking, this electron is under an *acceleration* v^2/r towards the nucleus and under a *Coulomb force* of $e^2/4\pi\epsilon_0 r^2$, so for stability,

$$\frac{e^2}{4\pi r^2 \epsilon_0} = \frac{mv^2}{r} \quad (5)$$

where m is the mass of the electron and ϵ_0 is a constant (known as the *permittivity* of free space). Then, using equations (5) and (4),

$$r = \frac{4\pi\epsilon_0(mvr)^2}{me^2} = \frac{4\pi\epsilon_0(n\hbar)^2}{me^2} \quad (6)$$

gives the possible discrete orbit radii. When $n=1$, this gives the *Bohr radius*, $r \cong 0.53 \times 10^{-10}$ metres.

The allowed energy levels of the electron in Bohr's hydrogen atom are given by the sum of the kinetic and potential energies, and so, using equations (5) and (6), are given by

$$E_n = \frac{1}{2}mv^2 - \frac{e^2}{4\pi\epsilon_0 r} = \frac{-e^2}{2(4\pi\epsilon_0 r)} = \frac{-me^4}{2(4\pi\epsilon_0 n\hbar)^2} \quad (7)$$

It then follows from equation (7) that if an electron 'jumps' from orbit n_2 to n_1 , with $n_2 > n_1$, then the loss in energy is

$$\frac{me^4}{8\varepsilon_0^2 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

which Bohr supposed was radiated as a single photon. Using equation (1), it follows immediately that the corresponding frequency is

$$\nu(n_1, n_2) = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (8)$$

Formula (8) is the Balmer-Rydberg formula (3) for hydrogen, with

$$R = \frac{me^4}{8c\varepsilon_0^2 h^3}$$

This correspondence enabled the Rydberg constant to be determined from the known constants and the result agreed with the experimentally determined value for R .

Modifications to Bohr's theory followed in quick succession. Finer spectral lines were discovered and then Pieter Zeeman (1865-1943) showed that yet more lines appeared when the source of the spectrum was placed near a magnet. The extra spectral lines due to magnetic effects were eventually explained by Wolfgang Pauli (1900-1958) and independently by George Uhlenbeck (1900-1988) and Samuel Goudsmit (1902-1978). The explanation put forward was that the extra lines were due to hidden rotation, or *spin*, of the orbiting electron. I will return to the fundamental idea of *electron spin* in later lectures on POAMS. With these and other theoretical modifications, the full informational contents of spectral lines were more or less explained on the basis of the hidden motions of these invisible elementary particles.

4. The first stage in the development of Quantum Mechanics

The first stage in the development of Quantum Mechanics began with Planck's introduction of the quantisation of action and lasted until about 1925. Following Planck's and Einstein's pioneering work, it was soon

discovered that a *discrete structure* was persistent throughout *all physical phenomena*. Any physical property is *quantised*, the **quantum** being the minimum amount of any physical property involved in an interaction. On the macro-scale, physical measures become effectively continuous due to the *correspondence principle* of Quantum Mechanics: Quantum Mechanics reproduces classical physics in the limit of large quantum numbers.

For example, it follows, from the equivalence of mass and energy, discovered by Einstein (discussed in Lecture 9), that if energy is ultimately quantised, then so is *mass*. Since action and energy are quantised, it follows by the Planck relation (1), that *time* is ultimately quantised. This agrees with the fact that duration is recorded by means of the ‘ticks’ of atomic clocks, by the use of atomic oscillations.

From the holistic standpoint of Normal Realism, all fundamental measures of physical phenomena are interrelated. Hence, whenever a practical reduction in any arbitrary unit of such a measure is contemplated, we have to think of it as being carried out within the context of the whole relational network of such measures. It follows, therefore, that at some stage in the reductive analysis of such fundamental quantities, we are bound to reach a level at which, for a particular set of observations, all we may contemplate is the bare numerical relationship between the various measure-categories. It seems reasonable, therefore, to suppose that for each particular phenomenon, at some stage in the reductive analysis, the units of certain fundamental measures are optimal in the sense that further subdivisions of a particular unit will cause others to increase. The minimum values of these units of fundamental physical measures, with respect to the whole relational system, can be thought of as the natural, indivisible units of such measures. Then equations involving these physical measures are ultimately *Diophantine equations*, that is, equations in integers.

Starting with things at the empirical level of reality, that is, with things we actually *observe*, we may say that these ultimate parts of observational elements into which those things may be sensibly and/or instrumentally analysed, have no qualities or characteristics other than in the contexts of those observational wholes of which they are the analysed parts.

In the next lecture, I shall provide a brief resume of the second stage of Quantum Mechanics before discussing the discrete nature of light-energy further.

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